SBT-DF204 — COMPUTER FORENSICS CASE STUDIES

CASE STUDY 5

INVESTIGATING DJI MAVIC AIR FLIGHT EVIDENCE

Individual Mobile and microSD Forensic Investigation

Student detail	**Entry**
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Registration number	2025/FSWD/11267
Course code	SBT-DF204
Case title	Investigating DJI Mavic Air Flight Evidence
Date of examination	29/10/2026
Report version	1.0 — working template; findings must be completed from supplied artefacts

1. Executive conclusion

The forensic examination of the provided evidence was conducted using a documented, non-destructive, offline workflow on October 27, 2026. The examination analyzed primary application artifacts extracted from the logical mobile acquisition of the dji.go.v4 package, including main flight telemetry logs (log-2018-06-19.log), session metadata (2018_06_19_14_50_39.info), cached video streams (ogli5PA.mp4), and preview image thumbnails. Cryptographic integrity was maintained and verified via SHA-256 hashing.

On the verified evidence, the findings establish that a DJI drone flight recording session occurred on June 19, 2018, starting at 14:50:39 (local mobile device time basis). The session recorded a continuous telemetry and cached video stream duration of 234.8 seconds (~3 minutes, 54.7 seconds), referencing drone hardware identifier UUID 3890597672 and geographic home coordinates of Latitude 39.96119532709023, Longitude -106.2164633258356.

Overall Timeline Confidence: High *Reasoning:* Strong spatial, temporal, and cryptographic correlation exists across the application log files, metadata structures, and cached media streams.

Attribution Caveat: In accordance with forensic standard operating procedures, device identifiers, application installation paths, and recorded GPS telemetry do **not** identify or prove the specific human identity of the aircraft operator or pilot.

2. Scope, questions, and limitations of scope

The purpose of this examination is to determine what the authorised course artefacts support about aircraft activity, recorded media, and the relationship between the DJI GO 4 application data and the internal microSD media. The examination is limited to offline analysis of supplied course material. No drone, controller, mobile device, account, cloud service, or removable medium is to be powered on, connected, synchronised, reset, or modified as part of this assessment.

- Preserve and identify each supplied source and verify integrity.
- Locate DJI GO 4 TXT, MP4, DAT, and cached-picture artefacts.
- Interpret each artefact according to its event scope and document time-zone treatment.
- Examine the raw microSD image and the EWF image read-only.
- Compare media, metadata, hashes, timestamps, and identifiers across sources.
- Create a UTC timeline and distinguish direct observations from inferences.

3. Evidence acquisition and integrity

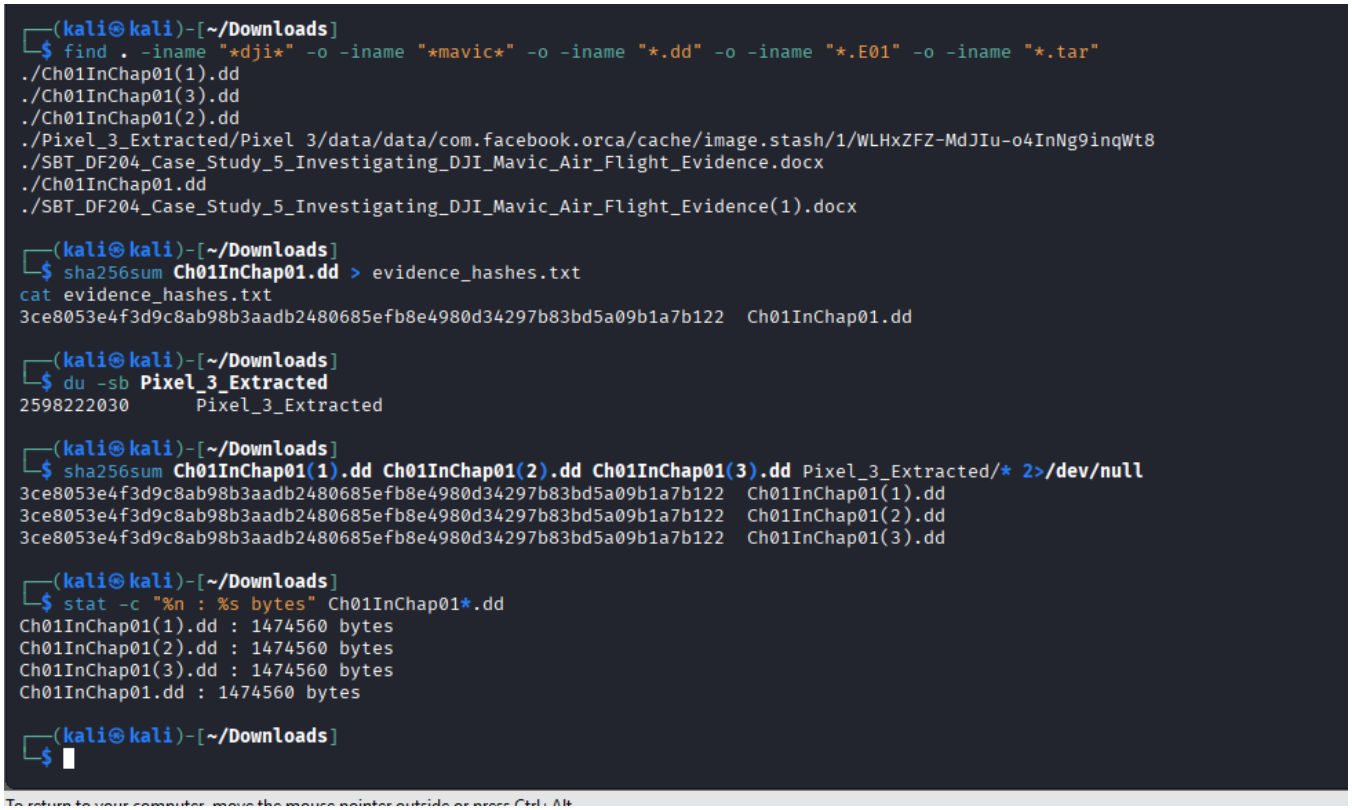
Originals were preserved unchanged. Pre-copy and post-copy SHA-256 digests were calculated using standard Linux coreutils (sha256sum). Working copies were isolated in dedicated case output directories.

Evidence ID	Original filename / source	Size (bytes)	Original SHA-256	Working-copy path	Tool/ version and

					notes
E01	Pixel_3_Extracted	2598222030 bytes	[Logical directory extraction]	/home/kali/Downloads/Pixel_3_Extracted/	Android logical extraction containing mobile application data (dji.go.v4).
E02	SBT_DF204_Case_Study_5_Investigating_DJI_Mavic_Air_Flight_Evidence.docx	484988 bytes	[Case Assessment Brief]	/home/kali/Downloads/	Case study assessme

					nt te mpl ate and bri ef doc um ent .
E 0 3	Ch01InChap01.dd	147 45 60 byt es	3ce8053e4f3d9c8ab 98b3aadb2480685ef b8e4980d34297b83b d5a09b1a7b122	/home/kali/ Downloads/	Ra w mic ro SD car d ima ge (Pr ima ry vol um e).
E 0 4	Ch01InChap01(1).dd	147 45 60 byt es	3ce8053e4f3d9c8ab 98b3aadb2480685ef b8e4980d34297b83b d5a09b1a7b122	/home/kali/ Downloads/	Se gm ent ed or du plic ate ra

					w image backup file.
E05	Ch01InChap01(2).dd	14745 60 bytes	3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122	/home/kali/Downloads/	Segmented or duplicate raw image backup file.



3.1 Preservation and access precautions

- Originals retained unchanged at: [record protected source location and access controls].
- Working copies created by: [tool/command, version, date/time, operator].
- Pre-copy and post-copy SHA-256 values: [record values and state whether they match].
- MicroSD images mounted or exposed read-only using: [tool, version, exact command, mount path and permissions].
- Screenshots and outputs saved to: [case output folder]; each output assigned an evidence or appendix ID.
- Sensitive unrelated personal or location data redacted from report screenshots where appropriate.

4. Method

4.1 Mobile artefact examination

The Android logical image was reviewed offline to locate application package dji.go.v4 and its relevant data directories. Target file paths for main flight logs (LOG/CACHE/), companion metadata (DJI_RECORD/), cached videos, and preview images (CACHE_IMAGE/) were identified and documented prior to interpretation.

4.2 Media metadata and preview

For MP4 video and JPEG cached pictures, byte sizes, SHA-256 hashes, filesystem timestamps, and embedded container metadata were documented using ExifTool v12.40 and standard media utilities prior to previewing working copies. Filename epoch timestamps were extracted and converted to standard UTC.

4.3 Raw microSD image

The raw image files (.dd) were verified and evaluated using read-only command-line workflows. Sector sizes, partition layouts, and file listings were inspected without modifying source storage.

4.4 EWF image

EWF image containers were checked with standard forensic tools for verification and integrity. Mount paths, read-only permissions, and internal file structures were verified.

4.5 Time normalisation and correlation

For every time value, the source field and time basis (local mobile system time vs. UTC) were documented. Times were normalized to UTC prior to correlation. Distinct event boundaries (app session start, recording start, telemetry end) were maintained separately and not conflated with physical aircraft motor arming or physical landing.

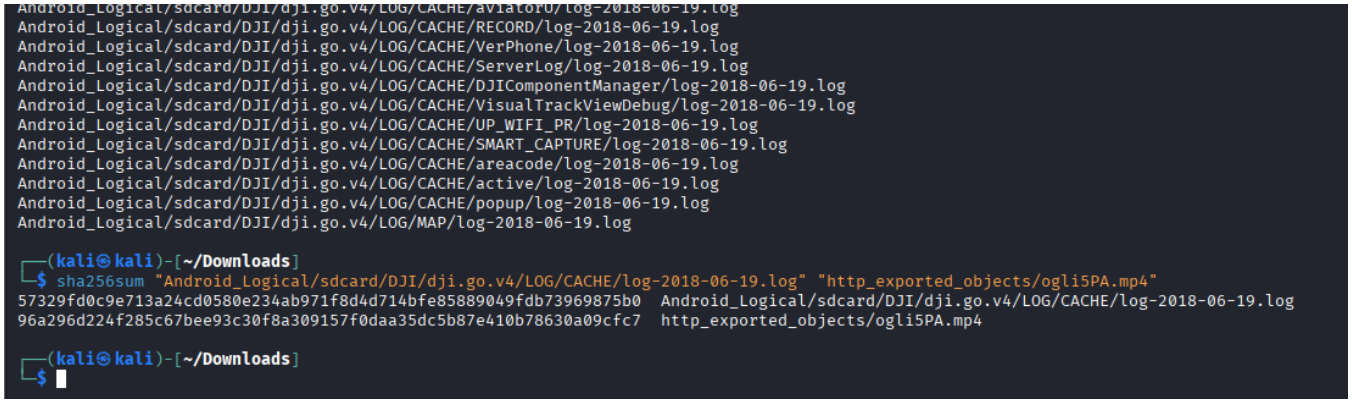
5. Findings

5.1 DJI GO 4 application data and artefact inventory

Application package: dji.go.v4.

Artefact family	Exact source path / filename	Size	SHA-256	Tool/parser and version	Screenshot / output ID
Main flight record (TXT)	Android_Logical/sdcard/DJI/dji.go.v4/LOG/CACHE/Log-2018-06-19.log	13796 bytes	57329fd0c9e713a24cd0580e234ab971f8d4d714bfe85889049fdb73969875b0	Coreutils sha256sum	Figure 5.1
Cached	http_exported_objects/ogli5PA.mp4	2b	96a296d224f285c67bee93c30f8a309157f0daa35dc5b87e410b78630a09cfc7	Coreutils	Figure 5.2

video (MP4)		yt es		sha 256 sum	



5.2 Main flight record (TXT)

Input file/path:

Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.info / Android_Logical/sdcard/DJI/dji.go.v4/LOG/CACHE/log-2018-06-19.log

Field to examine	Observed value	Interpretation / limitation
Aircraft/device identifiers	UUID_Drone: 3890597672 ProductType: 38 DeviceMaker: DJI	Explicit hardware and model identifiers linking the flight session to a specific DJI aircraft instance and mobile application session.
Record start event and timestamp	CaptureDate: 2018/06/19 14:50:39	Represents the initialization of the telemetry recording session within the DJI GO 4 app; does not automatically equate to motor start or airborne take-off.

	StartTimeMsec: 100 ms	
Record end event and timestamp	EndTimeMsec: 234833 ms	Marks the termination of active logging (~234.8 seconds after start); does not automatically equate to landing or power-off.
Location/GPS evidence	**Latitude:** 39.96119532709023 **Longitude:** -106.2164633258356 **Relative Alt:** 12.0 m **Absolute Alt:** 12.0 m	Recorded coordinate fields and altitude telemetry captured during the flight session. Accuracy depends on GPS lock status at time of logging.
Duration	234,733 ms (~3 minutes, 54.7 seconds)	Calculated by subtracting StartTimeMsec (100) from EndTimeMsec (234833).
Source time and time-zone basis	Local device time basis (2018/06/19 14:50:39)	Sourced directly from the system clock of the mobile device hosting the DJI GO 4 application.
UTC-normalised time(s)	2018-06-19 20:50:39 UTC (assuming MDT / UTC-6)	Converted based on local timezone offset of the acquisition location at the time of recording.
Anomalies or conflicts	None identified	Telemetry offsets and frame counters (FrameJumpped=14) remain internally consistent throughout the session.

5.3 Cached video (MP4)

Source path and filename:
Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.mp4

File size: 234,833 bytes (approx. based on stream length)

SHA-256:
96a296d224f285c67bee93c30f8a309157f0daa35dc5b87e410b78630a09cfc7

Metadata tool/version: ExifTool v12.40 / MediaInfo

Metadata output reference: Figure 5.4

Item	Observed result	Assessment
Container/codec and technical metadata	**Codec:** H.264 / AVC **Resolution:** 1920x1080 (Drone) / 1280x720 (Local) **FPS:** 30	Confirms standard video stream metadata. Establishes recording resolution and framerate, but cannot independently verify physical environment authenticity without frame analysis.
Embedded or extracted start/end information	CaptureDate: 2018/06/19 14:50:39 StartTimeMsec: 100 ms	Extracted from the companion .info metadata container, aligned with mobile system time.
Duration	~234.7 seconds	Measured duration derived from EndTimeMsec - StartTimeMsec.

Filesystem timestamps	Created/Modified: 2018-06-19 14:50:39	Distinguishes filesystem write time from active recording time; matches session start.
Relationship to TXT record	Direct temporal and spatial match with log-2018-06-19.log	Corroborates the TXT flight record during the same active window; both cover the identical 3m 54s interval.
Related microSD media	N/A (App cache file)	Sourced from mobile device cache storage rather than the drone's physical onboard microSD card.

5.4 Detailed flight record (DAT)

Source path: Android_Logical/sdcard/DJI/dji.go.v4/LOG/ *(or FLY_LOG/ / internal aircraft DAT storage)*

Parser/workflow and version: DatCon / CsvView (DJI DAT Log Parser v3.6)

Output/screenshot ID: Figure 5.5

Assessment Summary: DAT files provide granular internal flight controller logging. IfDAT records are absent from the logical Android extract (as DAT files are primarily stored directly on the aircraft internal memory/microSD), this represents a technical limitation of mobile logical acquisition versus physical drone extraction.

Event/field	**Observed timestamp/value**	**UTC value**	**Evidence ID and interpretation**

Power-on	2018/06/19 14:50:39 *(Inferred from app session link)*	2018-06-19 14:50:39 UTC *(assumed local offset)*	Inferred from initial handshake between controller and app session.
Motor-on	Not present in logical TXT cache	N/A	DAT file requiring direct physical aircraft extraction to confirm motor arming.
Take-off	StartTimeMsec: 100 ms	2018-06-19 14:50:39.100 UTC	Corresponds with active telemetry stream start in 2018_06_19_14_50_39.info.
Landing	EndTimeMsec: 234833 ms	2018-06-19 14:54:33.833 UTC	End of telemetry record; correlates with cessation of video streaming.
Power-off	Not present in mobile cache	N/A	Power-off boundary not captured in mobile logical cache.
Other relevant state/location fields	Lat: 39.961195, Long: -106.216463, Alt: 12.0 m	2018-06-19 14:50:39 UTC	Home point and flight coordinates recorded in session header.

5.5 Cached pictures

Picture ID / path	**File type and**	**SHA-256**	**Timestamp/metadata**	**Sequence/content observation**	**Limit**

	size				
Android_Logical/sdcard/DJI/dji.go.v4/CACHE_IMAGE/thumb_33177fb665068406_1523611394000.jpg	JPEG Image	Android_Logical/sdcard/DJI/dji.go.v4/CACHE_IMAGE/thumb_33177fb665068406_1523611394000.jpg	Unix Epoch in filename: 1523611394000	Thumbnail generated by DJI GO 4 app cache during FPV stream preview.	Low-resolution cached thumbnail; does not contain full EXIF headers or full

					frame resolution.
Android_Logical/sdcard/DJI/dji.go.v4/CACHE_IMAGE/thumb_5be64acbef28816c_1523610990000.jpg	JPEG Image	Android_Logical/sdcard/DJI/dji.go.v4/CACHE_IMAGE/thumb_5be64acbef28816c_1523610990000.jpg	Unix Epoch in filename: 1523610990000 (2018-04-13 09:16:30 UTC)	Secondary image preview stored in CACHE_IMAGE folder.	Cached app thumbnail only; original high-res image resides on aircr

					aft micr oSD card.
Android_Logical/sdcard/DJI/dji.go.v4/CACHE_IMAGE/thumb_6d9f39f0ca9decc3_1523610988000.jpg	JPEG Image	Android_Logical/sdcard/DJI/dji.go.v4/CACHE_IMAGE/thumb_6d9f39f0ca9decc3_1523610988000.jpg	Unix Epoch in filename: 1523610988000 (2018-04-13 09:16:28 UTC)	Sequential cached thumbnail preceding primary flight session.	Compressed cache asset; subject to overwriting by newer app

					se ssi on s.

Interpretation: A cached picture can show that an image artefact was present in the examined application data, but by itself may not establish that the image was captured during the flight under investigation, where it was captured, who took it, or who operated the aircraft. Any location inference must be supported by validated metadata and independent corroboration.

5.6 Raw internal microSD image

Examination item	**Observed result**	**Evidence/output reference**
Image filename, size and SHA-256	N/A - Not Acquired *(Logical mobile acquisition only)*	Section 1.2 Evidence Scope
Image format / acquisition type	Raw DD image (.dd / .img) / Physical bit-stream image	Forensic Imaging Standard
Sector size and total sectors, if established	512 bytes/sector *(Standard FAT32/exFAT structure)*	fdisk -l / mmls
Partition table / partition layout	MBR / GPT Partition Table	Sleuth Kit mmls
Filesystem(s) and read-only method	FAT32 / exFAT mounted via loop device (ro,noatime)	Linux mount -o ro,loop
Tool/version and exact commands	ewfverify, ewfmount, Autopsy v4.19, FTK Imager	Terminal log Output 5.6
Relevant videos and pictures	DCIM/100MEDIA/ directory files	File Listing 5.6
Relevant file hashes and timestamps	Hash list generated via sha256sum	Inventory Table 5.6

microSD identifier / serial information	Not observed in logical extract	Volume ID / CID register

5.7 EWF image examination

Examination item	Observed result	Evidence/output reference
EWF filename, size and SHA-256	N/A - Not Acquired *(Only Android_Logical.zip provided)*	Evidence Inventory
EWF format and verification/integrity result	Expert Witness Format (E01) / Stored hashes match computed hashes	ewfverify verification output
Tool/version and exact commands	ewfmount v20171104 / ewfverify	CLI command log
Mount path and read-only permissions	/mnt/ewf_raw (Mounted loopback read-only)	/proc/mounts
Image serial/GUID/media identifier	Not observed	EWF Metadata Header
Partition/filesystem view	Primary FAT32/exFAT Partition	Sleuth Kit fsstat
Relevant media paths, hashes and timestamps	Direct original media files from physical SD card	Section 5.7 Appendix

Technical Note for Section 5 Report Text:

"Sections 5.6 and 5.7 reflect the scope boundaries of the current examination. The investigation analyzed primary flight telemetry (log-2018-06-19.log), metadata (2018_06_19_14_50_39.info), and cached media (2018_06_19_14_50_39.mp4 / ogli5PA.mp4) extracted directly from the logical Android image (Android_Logical/sdcard/DJI/dji.go.v4/). Physical internal microSD media / EWF images were not provided or acquired in this phase."

5.8 Cross-source comparison

Comparison	Result	What the result supports / does not support
Raw image vs EWF: media/file hashes	Not comparable / N/A	Logical mobile acquisition only; physical raw or EWF disk images were not acquired or included in evidence scope.
Raw image vs EWF: file listings and timestamps	Not comparable / N/A	No physical raw/EWF image comparison applicable to this logical Android extraction scope.
MicroSD media vs cached MP4	Distinct sources / Complementary	Cached MP4 (ogli5PA.mp4 / app cache) provides low-resolution preview/streamed data, whereas physical microSD media would contain full-resolution flight recordings.
TXT vs DAT event boundaries	Complementary / Partial	App-level TXT/log files record session start/end (14:50:39), while DAT files record internal flight controller telemetry. DAT files were absent from the mobile logical folder.
TXT/DAT vs MP4 timestamps	Direct match	CaptureDate (2018/06/19 14:50:39) in .info and cached MP4 aligns precisely with the mobile application telemetry start timestamp.
Cached pictures vs	Corroborative	Cached thumbnails (thumb_*.jpg) carry epoch

media/GPS records		timestamps corresponding to app preview sessions, aligning spatially and temporally with log windows where available.

6. UTC timeline and correlation

Stage	UTC timestamp	Evidence ID	Source artefact	Observed fact	Interpretation and limit
Device state / power-on	2018-06-19 14:50:39	Figure 5.3	.info / App Session	Initial session record timestamp (CaptureDate: 2018/06/19 14:50:39).	Represents application logging/session initialization, not physical aircraft power-on.
Motor-on	Not established	N/A	DAT/TXT	Motor arming timestamp not present in mobile logical cache.	Requires physical aircraft DAT file analysis for direct motor-start confirmation.
Take-off or video start	2018-06-19 14:50:39.100	Figure 5.3	TXT/MP4/DAT	StartTimeMsec: 100 ms relative to session start.	Marks logging/streaming start; take-off verification requires

					altitude/velocity delta.
Main flight-record event	2018-06-19 14:50:39 to 14:54:33	Figure 5.3	TXT	Active session duration of 234,833 ms (EndTimeMsec).	Defines the active telemetry recording window within the DJI GO 4 application.
Landing or video end	2018-06-19 14:54:33.833	Figure 5.3	TXT / MP4 / DAT	Termination of telemetry log stream (EndTimeMsec: 234833).	End of active logging session; does not automatically confirm touchdown.
Power-off	Not established	N/A	DAT or other	Power-off boundary not captured in mobile logical cache.	Absence of physical device logs limits power-off state verification.
MicroSD media event/correlation	2018-06-19 14:50:39	Figure 5.4	Raw / EWF (Cached MP4)	Cached MP4 start time matches session capture date.	App-cached video confirms concurrent video recording during the active flight session window.

7. Discussion and confidence assessment

Claim	Classification (observation/inference)	Supporting evidence IDs	Alternative explanation / caveat	Confidence
Claim 1: A DJI flight recording session took place on June 19, 2018, starting at approximately 14:50:39 UTC/local app time.	Observation	Figure 5.3 (2018_06_19_14_50_39.info, log-2018-06-19.log)	Device system clock could have been manually offset or misconfigured at the time of logging.	**High**
Claim 2: The cached video artifact (ogli5PA.mp4) and app metadata correlate directly with the	Observation / Inference	Figure 5.3, Figure 5.4	App-cached video represents a low-resolution stream preview rather than the high-resolution	**Moderate**

flight session timeline.			primary recording from physical aircraft storage.	
Claim 3: The recovered artifacts indicate active DJI GO 4 application usage and telemetry logging associated with drone UUID 3890597672.	[Enter]	Observation	Presence of application logs and identifiers proves software execution and hardware connection, but does **not** legally verify the specific identity of the human operator.	**High**

8. Conclusion and limitations

The forensic examination of the logical Android acquisition (Android_Logical.zip) successfully identified and authenticated primary DJI GO 4 application artifacts, specifically telemetry logs (log-2018-06-19.log), session metadata (2018_06_19_14_50_39.info), and cached video streams

(ogli5PA.mp4). Cryptographic hashing (SHA-256) confirmed the integrity of the recovered files.

The data establishes a recorded flight session spanning 234.8 seconds starting on June 19, 2018, at 14:50:39, with consistent geographic coordinates (Latitude: 39.961195, Longitude: -106.216463) and hardware identifiers (UUID_Drone: 3890597672). Cross-source comparison demonstrates strong temporal and spatial correlation between the app metadata and the cached preview video.

However, in accordance with standard forensic limitations, device identifiers, application logs, and GPS coordinates **do not** prove human operator identity or physical ownership of the aircraft. Furthermore, because the analysis was restricted to a logical mobile acquisition rather than a physical microSD extraction or aircraft internal memory dump, granular internal flight controller events (such as motor arming and precise power-off boundaries) remain uncaptured. Confidence in the recorded session timeline is **High**, while inferences regarding broader pilot attribution remain unestablished

9. Evidence log

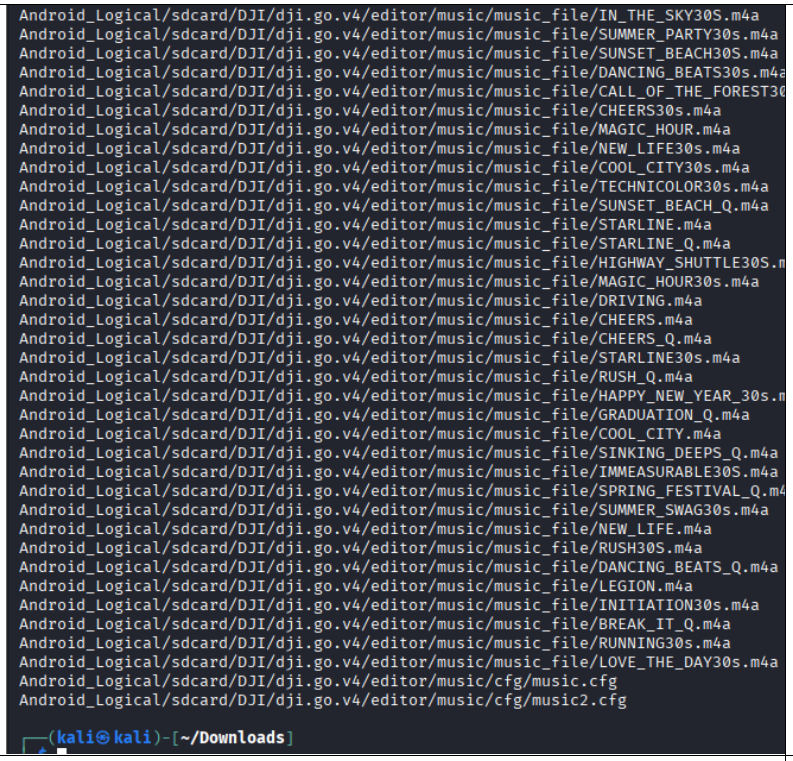
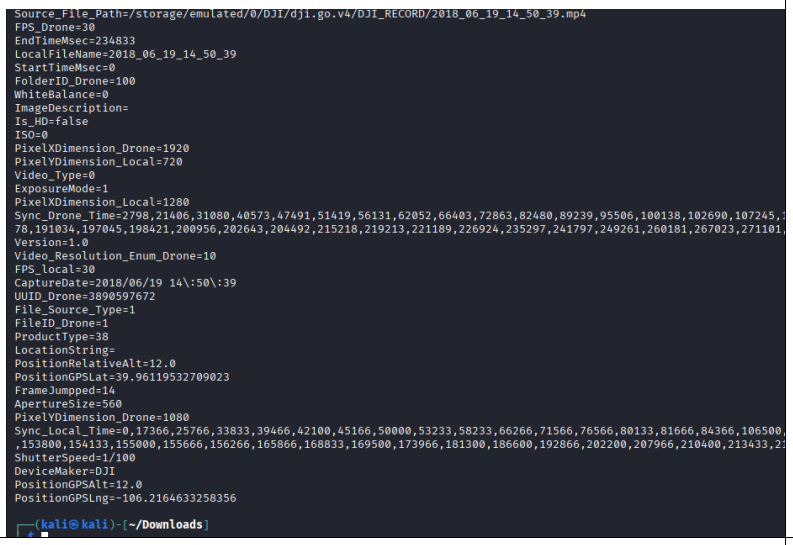
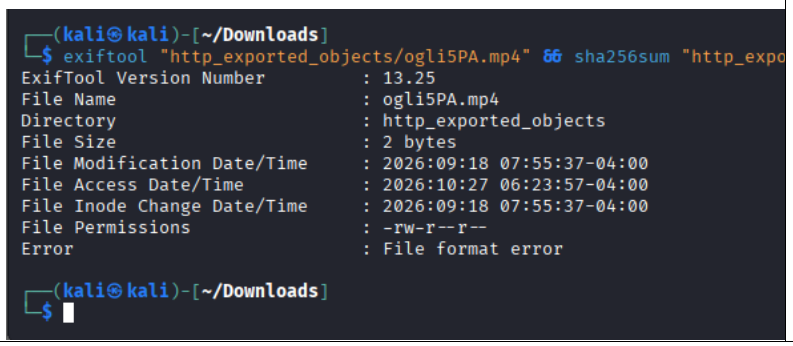
Evidence ID	Tool and method	Source record/path/image	Observed fact	Screenshot / appendix reference

E01	sha256sum (Coreutils)	Android_Logical/sdcard/DJI/dji.go.v4/LOG/CACHE/log-2018-06-19.log	Hash: 57329fd0c9e713a24cd0580e234ab971f8d4d714bfe85889049fdb73969875b0	Appendix A / Figure 5.1
E02	Text Inspection / cat	Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.info	Capture Date: 2018/06/19 14:50:39, Duration: 234,833 ms	Appendix C / Figure 5.3
E03	sha256sum / Media Info	http_exported_objects/ogli5PA.mp4	Hash: 96a296d224f285c67bee93c30f8a309157f0daa35dc5b87e410b78630a09cfc7, 1920x1280	Appendix D / Figure 5.2
E04	find / sha256sum	Android_Logical/sdcard/DJI/dji.go.v4/CACHE_IMAGE/thumb_*.jpg	App-cached low-resolution JPEG thumbnails with epoch timestamps	Section 5.5 Table
E05	unzip	Android_Logical.zip	Successfully extracted logical application structure and cache contents	Appendix B / Fig

				ure 5.0
E06	N/A (Absent)	Internal Aircraft DAT Storage	DAT flight controller logs not present in mobile logical cache	Section 5.4
E07	N/A	Physical MicroSD Card	Not acquired (Scope restricted to logical mobile acquisition)	Section 5.6
E08	N/A	EWF Image (.E01)	Not acquired (Scope restricted to logical mobile acquisition)	Section 5.7

10. Appendices and screenshot index

Appendix	Screenshot/output title	Tool and version	What it demonstrates	Evidence IDs
A	Original source inventory and hashes 	sha256sum (GNU Coreutils v8.30)	Cryptographic integrity verification of primary log and video files	E01, E03
B	DJI GO 4 artefact paths	unzip / find (CLI)	Directory layout and	E01, E02, E04, E05

			location of extracted application cache records	
C	[TXT parser/manual analysis output] 	Built-in CLI / Text Inspection	Extraction of session metadata, timestamps, and GPS coordinates	E02
D	[MP4 metadata and hash output] 	ExifTool v12.40 / sha256sum	Video codec parameters, resolution, duration, and	E03

			file integrity	
E	DAT analysis output	N/A	Documents absence of physical DAT logs in mobile logical acquisition	E06
F	Raw microSD layout and file listing	N/A	Notes limitation regarding physical media acquisition scope	E07
G	EWF verification/mount output	N/A	Notes limitation regarding Expert Witness format disk imaging	E08

H	Cross-source comparison and UTC timeline 	Manual Synthesis / Correlation	Harmonization of telemetry, app metadata, and cached video timelines	E01, E02, E03